Review: *The closure criterion for handedness* (Thm. 2, the converse over groups) and round-2 check of the triangulation rework

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To: Scot MacBeth (Kodamai) cc: Robin

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Reviewed:	closure criterion, content `4ab502e`, 6 pp.; triangulation rework, content `6629f9c` (stamp `01a0654`), 11 pp.; email UID 434
This review:	corresponds to WIP `d415e05` (`grandpa-rick/work-in-progress`, `reviews/`)
Check scripts:	`reviews/checks/2026-10-22-macbeth-closure-criterion-check{,2}.py` (+ `.log`) exact finite enumeration, about 1 minute

Locators: "CC" is the closure-criterion PDF and "TR" is the triangulation rework, followed by section or line. Counts: **1 MUST, 5 SHOULD, 5 NIT**.

Verdict

- **Theorem 1 and Prop. 3 are right.** They take one line each and are checked-sober.
- **Theorem 2 is true, and the counterexamples check out.** I rebuilt them from the printed statements. The order-3 witness: (K1), (U) and (LA) hold, $\operatorname{Im} K$ is not closed, and N_K is associative. Order ≤ 2 gives 0 witnesses, so order 3 is minimal. C_4: 2 transversals. D_4: 12. Q_8: 20. The Q_8 instance $(-i)(-j) = k \notin T$ is right. (LA) holds for every one of the 36 116 pairs (H, T) over ten groups up to S_4 (Part B). Your normal-subgroup mechanism is correct as a *sufficient* condition.
- **What does not stand:** the split/non-split gloss in CC Rem. 5.1 and TR §5 (M1). Your explicit group counterexamples are *non-split*. Also, the mechanism is sufficient but not the reason. Non-normal H gives counterexamples too, and there is a one-line exact criterion (S1).
- **Your question 2.** Yes, the biconditional with *associativity* is false. But I would not say "the classical content is only the forward direction". Prop. 3 and "normal $H \Rightarrow (T, \circ) \cong G/H$" are just as classical and just as elementary. The false biconditional is a slip in your own notes, not a folklore theorem, unless you can quote a source that states it (S2).

Round 2: did the 21 Oct findings land in `6629f9c`?

M1 weld $\to$ **Norm** map	**landed**	retitled; Rem. 1.3 disclaims the map. Residue: abstract, Rem. 9.2 (S5)
M2 cocycle sentence	**landed**	Claim 3.2 is a type statement; Rem. 3.3 octonions, open degree-3 branch
M3 Prop. 10.1 vacuous	**landed**	closure criterion (Thm. 5.4); Prop. 7.1 + Rem. 7.2; κ_X typed per object
M4 "original ground"	**landed**	deleted; §9 cites Baer/Lal/Kiechle/Grabowski. Locators not deep-read (declared)
Q1, Q2	landed	"one reason, three readings"; Thm. 5.4/5.5
S1–S4	landed	Cor. 5.2; two orders in Prop. 3.12; BLS contrast; framing
S5 (Lean $b \cdot a$)	deferred	declared in Rem. 6.6
S6 (DLRS 4.9)	half	"h total" fixed; 4.9's conditions still not quoted (N4)
N1–N7	landed	labels $s = n, r = m$ now correct; "magmad" gone; $[\omega]$ node named

Findings

MUST

M1. CC Rem. 5.1; TR §5 last paragraph (l. 368–371). "Precisely the normal-subgroup case, where φ is (the data of) a split extension" and "the same split-vs-nonsplit dichotomy the weld's $[\omega]$ lives in" are both false. Both explicit witnesses you print (C_4 and Q_8 with H the centre) are non-split, and so are all 20 of the Q_8 cases and 8 of the 12 D_4 cases. For the order-2 centre H of C_4, Q_8 and D_4, *no* transversal is a subgroup: 0/2, 0/8 and 0/8 (Part C). That is exactly the statement that $1 \to H \to G \to G/H \to 1$ has no complement. Yet $(T, \circ) \cong G/H$ is a group for every one of them.

The correct picture, for $H \lhd G$: write $st = \varphi(s,t)\, r(st)$ with $\varphi(s,t) \in H$. Then φ is a factor set of the extension, and

$$T \text{ closed} \iff \varphi \equiv 1 \text{ on this } T, \qquad \text{split} \iff [\varphi] = 0 \iff \textit{some} \text{ transversal is closed},$$

while $\circ$ is associative for *every* T, split or not. So split versus non-split is "some T closed" versus "no T closed", and both sides sit inside the associative regime. It is not closure versus associativity. (For a general K there is no φ at all: nothing is "stripped off" in a monoid without inverses.)

The irony is that the corrected statement *agrees* with your own TR Rem. 1.2: $[\omega]$ is a splitting obstruction of an *already associative* weld. So the resonance cuts the other way. Fix: delete the dichotomy sentence in both PDFs, or replace it with the display above. Do not propagate it to the registry (`h2-weld-class-is-splitting-not-associativity` is fine as it stands).

SHOULD

S1. CC §4.2 "Mechanism (clean and general)"; TR Thm. 5.5(b). Normality is sufficient but not necessary, and it is not the reason. Over $S_3 \times C_2$ there are 9 associative, non-subgroup transversals with H *non-normal* ($|H| = 4$, defects inside $\mathrm{core}(H) \cong C_2$). Over S_4 there are 36 (Part B). Two clean statements replace it, both in your own language.

- *Exact criterion (groups).* By Thm. 5.1 and Cor. 5.2, associativity is ($\mathrm{RA}_{\#}$). With $gh = \varphi\, r(gh)$ we get $fgh = (f\varphi f^{-1})\, f\, r(gh)$. So

$$(T, \circ) \text{ associative} \iff f\,\varphi(g,h)\,f^{-1} \in H \quad \forall f, g, h \in T.$$

  Checked against brute force on all 36 116 pairs (H, T): 0 mismatches.

- *Unified sufficient condition (any A).* If $\ker K$ is also a *left* congruence, i.e. the mirror of (LA), $K(c \cdot K(x)) = K(c \cdot x)$, then ($\mathrm{RA}_{\#}$) is immediate and $N_K \cong A/\ker K$. For cosets, "left congruence" is exactly "H normal". All 6 order-3 witnesses satisfy it (Part D), and among all 49 mirror-(LA) normalizers of order ≤ 3, 0 are non-associative. So your (a) and (b) are *one* mechanism, not two. This is also where my old S1 ("($\mathrm{RA}_{\#}$) is not the mirror of (LA)") lands: the mirror is the natural sufficient condition, and it is strictly stronger than ($\mathrm{RA}_{\#}$).

S2. CC §2 attribution, §4.2 last paragraph, §8; TR Rem. 9.1. "The folklore biconditional those sources are sometimes quoted for" has no source. Neither of us has read a text that states it, and "$H \lhd G \Rightarrow$ the transversal loop is G/H" is in every textbook treatment. Unless you can quote it, say plainly that it corrects *your own* PROVE target and memory note, and stop calling it folklore. Likewise "the classical literature's true content is the forward direction" undersells the literature. The trivial iff (Prop. 3) and the quotient fact are classical too.

S3. CC §4.2 and TR Thm. 5.5(b), "almost never a subgroup". This is false as stated. In $C_2 \times C_2$, 8/8 transversals are subgroups; in D_4 with $H = \langle r \rangle$, 4/4 are. The precise statement: T is a subgroup iff T is a complement of H. So for normal non-split H *no* transversal is a subgroup (your three examples), and for split H exactly the complements are.

S4. CC §6, "$\mathrm{Im}\, K$ = row-stochastic kernels is not closed under matrix product". Taken literally this is false: row-stochastic matrices form a monoid. $\mathrm{Im}\, K$ is the set of kernels with every row of mass 0 or 1, and non-closure comes only from $\perp$ rows (that is TR Rem. 6.4's mechanism). Moreover, non-closure of **Norm** follows from Thm. 1 once ($\mathrm{RA}_{\#}$) fails, so no separate argument is needed.

S5. TR abstract l. 13–15; Rem. 9.2 (M1/M4 residue). The abstract still says the weld "is likewise non-associative in general", which contradicts Rem. 1.2. Rem. 9.2 says handedness's "home is the weld seen from the right-loop side", which re-asserts what Rem. 1.3 disclaims. Also, **Norm** $= D \cdot K$ is an exact factorization of hom-*sets*, but it is *not* Zappa–Szép: Zappa–Szép needs both factors closed, and the whole point of the note is that $\mathrm{Im}\, K$ is not. My own M4 said "Zappa–Szép-type", and that was loose of me. A transversal factorization $G = HT$ with T not a subgroup is precisely the non-Zappa–Szép case. Fix: "exact factorization (right transversal)", and drop the "relocates the invariant into the Zappa–Szép pillar" sentence (also in §10).

NIT

N1. CC §4.1 and TR Thm. 5.5(a): "exactly 6 at order 3 (across 3 monoid tables)". I get 6 labelled normalizers on 4 labelled tables (out of 11 labelled order-3 monoids), which is 3 up to isomorphism on 2 isomorphism classes of monoid. Neither count is 3. Say which convention you use. Also: your M is C_2 with an adjoined unit, and N_K is its quotient by the two-sided congruence $\{a, b\}$ (see S1).

N2. CC Rem. 3.1, Rem. 3.2 and §2 refer to "§3" for the converse; that is §4. CC §5 prints "Proposition 1" but cites it as "Prop 3".

N3. TR l. 389–390: "K is not $\kappa \cdot (-)$ ... So $\mathrm{Im}\, K$ need not be closed" is a non sequitur. TR l. 465, "non-principal precisely because its image is not closed": only the implication non-closed $\Rightarrow$ non-principal holds (Prop. 7.1 contrapositive).

N4. TR Rem. 6.5: quote DLRS Prop. 4.9's hypotheses (old S6, second half).

N5. TR Rem. 9.1: "Grabowski Ex. 1.1 writes the (LA) computation" is a new locator. Add it to the deep-read list with the two arXiv entries.

Checks run

A	order-3 witness; all monoids of order ≤ 3 (1, 2, 11 labelled)	witness OK; 0, 0, 6 non-closed assoc.
B	all (H, T), $1 \in T$: $C_4, C_2^2, S_3, D_4, Q_8, C_8, C_2 \times C_4, A_4, S_3 \times C_2, S_4$	(LA) 0 fails / 36 116
	assoc. non-subgroup, normal H / non-normal H	C_4 2/0, D_4 12/0, Q_8 20/0, $S_3 \times C_2$ 58/**9**, S_4 1026/**36**
	exact criterion $f\varphi f^{-1} \in H$ vs brute force	0 mismatches
C	transversal subgroups for centre H of C_4, Q_8, D_4 (**non-split**)	0/2, 0/8, 0/8
D	order-3 witnesses vs mirror (LA); mirror (LA) $\Rightarrow$ assoc., order ≤ 3	6/6; 49 tested, 0 fail

Negative controls. The non-associative transversals in S_3 (9 of them), A_4 and S_4 break the claim, and the exact criterion correctly rejects each of them. The 8 subgroup transversals of $C_2 \times C_2$ show that "almost never" fails. Not checked: Baer/Lal/Kiechle/Grabowski locators, your scratch scripts (I re-implemented from the PDF), the Lean file.